TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Next-Generation Physics-Informed Intelligent Platform for Quantitative Medical Imaging			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Biomedical			
	Medium Category	Small Category	Medium Category	Small Category
	Medical Devices	*Medical Imaging Devices*		
Concept of Proposed Technology	1. Core Concept The proposed technology aims to develop a next-generation intelligent medical imaging platform for PET, CT, and MRI that integrates physics-based modeling, advanced computational methods, and artificial intelligence (AI) to significantly improve image quality, quantitative accuracy, and clinical reliability. The core concept is a co-development framework in which cutting-edge image generation, correction, and processing algorithms are jointly developed with leading scientists at Mass General Brigham (MGB), and subsequently implemented and commercialized by Korean small and medium-sized enterprises (SMEs). The developed technologies will be clinically validated using large-scale, real-world imaging datasets from MGB-affiliated hospitals, ensuring robustness, generalizability, and immediate clinical relevance. Rather than focusing solely on incremental image enhancement, this project emphasizes clinically meaningful improvements, such as accurate quantification, reproducibility across scanners, and workflow-integrated deployment, enabling translation from research algorithms to deployable medical imaging solutions. 2. Potential Functions			

2.1 Advanced Image Generation

- Reconstruction of high-quality PET, CT, and MRI images from low-dose, fast-acquisition, or incomplete data
- AI-based synthesis of standard-quality images from reduced radiation dose PET/CT or accelerated MRI acquisitions
- Preservation of diagnostically and quantitatively relevant features critical for clinical interpretation

2.2 Physics-Informed Image Correction

- Hybrid correction methods combining imaging physics models and AI for:
  - PET attenuation, scatter, motion, and time-of-flight (TOF) correction
  - CT metal artifact and beam-hardening correction
  - MRI motion and field inhomogeneity correction
- Improvement of quantitative reliability while maintaining physical interpretability and regulatory feasibility

2.3 Quantitative Image Processing and Analysis

- Robust voxel-level and organ-level quantitative analysis across PET, CT, and MRI
- Multi-modality image registration and fusion for integrated anatomical–functional assessment
- Automated segmentation and biomarker extraction to support clinical decision-making

2.4 Clinical Validation Framework

- Performance evaluation using large-scale, heterogeneous clinical datasets from MGB
- Validation based on clinically relevant endpoints, including diagnostic accuracy, lesion detectability, and quantitative reproducibility
- Establishment of standardized evaluation protocols applicable across institutions and imaging platforms

3. Potential Uses and Applications

3.1 Clinical Diagnosis and Treatment Planning

- Enhanced diagnostic confidence through improved image quality and quantitative accuracy
- Support for precision oncology, neurology, and cardiovascular imaging
- Improved treatment planning and response assessment, particularly in molecular imaging–based therapies

3.2 Personalized and Theranostic Medicine

- Reliable quantitative PET imaging for patient-specific dosimetry and therapy optimization
- Integration into theranostics workflows for internal radionuclide therapy and response monitoring

3.3 Medical Device and Software Commercialization

- Core algorithms embedded into:
  - Imaging scanners
  - Reconstruction workstations
  - PACS and AI-assisted clinical software
- Accelerated global market entry supported by validation in a world-leading clinical environment

3.4 Research and Clinical Trial Enablement

	• Standardized imaging pipelines for multi-center clinical trials • Reduction of imaging variability, enabling more efficient and cost-effective trials
Rationale for Joint R&D Project	Recent advances in AI-based medical imaging have demonstrated great potential, yet many technologies remain confined to research settings and have not been effectively translated into routine clinical practice. Key barriers include limited access to large-scale, high-quality clinical datasets, insufficient prospective validation, and weak integration with real-world imaging workflows. Korean small and medium-sized enterprises (SMEs), while demonstrating strong technological implementation and commercialization capabilities, face inherent limitations in gaining access to diverse imaging data and conducting validation studies at internationally recognized clinical institutions. To overcome these structural challenges, strategic collaboration with Mass General Brigham (MGB) is essential. MGB provides a world-leading clinical ecosystem, housing extensive multimodal PET/CT/MRI datasets and advanced expertise in image reconstruction, quantitative analysis, and AI-driven clinical interpretation. This partnership enables co-development of clinically validated AI imaging technologies through the integration of complementary strengths—scientific innovation from MGB and engineering and deployment expertise from Korean SMEs.
Key Words	• Physics-informed AI Medical Imaging • Quantitative PET/CT/MRI Imaging • Low-dose & Accelerated Image Reconstruction • Commercial-grade Imaging Platform

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)

R&D Objectives	**1. AI-Based Image Generation for Low-Dose and Fast Imaging**
	• Reconstruction of diagnostic- and quantitative-quality images from low-dose PET/CT and accelerated MRI acquisitions
	• AI models trained on paired clinical datasets to synthesize standard-quality images while preserving clinically relevant features
	• Technology designed to reduce radiation exposure, shorten scan time, and improve patient comfort without compromising diagnostic performance
	
	2. Physics-Informed Image Correction Algorithms
	• Hybrid algorithms combining imaging physics models with deep learning for robust and interpretable correction
	• Targeted developments include:
	  ○ PET attenuation, scatter, motion, and time-of-flight (TOF) correction
	  ○ CT metal artifact and beam hardening correction
	  ○ MRI motion and field inhomogeneity correction
	• Emphasis on quantitative reliability and regulatory feasibility
	
	3. Quantitative Imaging and Multi-Modality Data Fusion
	• Accurate voxel-level and organ-level quantification for PET, CT, and MRI
	• Automated segmentation and feature extraction for lesions and organs
	• Integrated PET/CT/MRI registration and fusion enabling combined anatomical–functional analysis
	
	4. AI-Driven Clinical Decision Support Modules
	• Extraction of quantitative imaging biomarkers to support diagnosis, treatment planning, and response assessment
	• Tools designed for seamless integration into clinical workflows and imaging workstations
	• Focus on transparency and consistency to support clinical trust and adoption
	
	5. Clinical Validation and Benchmarking Framework
	• Standardized validation pipelines using large-scale, real-world clinical datasets from MGB
	• Performance evaluation based on clinically meaningful endpoints such as diagnostic accuracy, lesion detectability, and reproducibility
	• Establishment of reference benchmarks applicable across institutions and imaging platforms
	
	6. System Integration and Commercialization-Ready Implementation
	• Optimization of algorithms for real-time or near–real-time execution
	• Integration into imaging devices, reconstruction engines, and clinical software platforms
	• Design aligned with international regulatory and commercialization requirements			

R&D Target Specifications	• Achieve ≥ 20% improvement in signal-to-noise ratio (SNR) and ≤ 5% quantitative bias in reconstructed PET/CT and MRI images. (ground truth: regular full scan) • Enable ≥ 50% reduction in radiation dose and ≥ 40% reduction in scan time without loss of diagnostic accuracy. • Implement physics-informed correction algorithms achieving ≥ 80% artifact reduction and < 3% quantitative deviation across modalities. • Attain ≥ 0.8 Dice coefficient for organ/lesion segmentation and ≤ 3 mm multimodal			

	registration error. • Deliver real-time or near–real-time image reconstruction within ≤ 3 minute per volume.
R&D Plan	**1. Overall Research Approach** The project will adopt a phased, milestone-driven R&D strategy to systematically progress from feasibility analysis to full-scale development and real-world clinical validation within a three-year period. An initial six-month *Planning and Pilot Phase* will establish core frameworks—finalizing clinical requirements, defining data governance, and validating feasibility using pilot datasets from Mass General Brigham (MGB). Subsequent phases will focus on three objectives: 1. Developing high-performance multimodal AI algorithms and fusion models. 2. Implementing clinically deployable, optimized software systems. 3. Conducting rigorous multi-center clinical validation and benchmarking to ensure FDA-ready performance and manufacturability. The approach emphasizes quantitative validation, clinical integration, and commercialization-readiness by the final project stage. **2. Key Tasks and Methodologies (Work Packages)** Phase 0 / WP0. Planning and Pilot Study (Months 1–6) *Objective:* Establish the foundation for joint R&D operations and technical feasibility. • Finalize clinical use cases and target performance metrics. • Define governance structures, IP frameworks, and data access protocols (including federated learning). • Select pilot PET/CT/MRI datasets from MGB and validate preliminary model feasibility. *Milestones:* • M0.1: Joint research and governance framework established. • M0.2: Roles, IP, and data governance finalized. WP1. Core Algorithm Development (Months 7–24) *Objective:* Develop advanced multimodal AI algorithms for image generation, correction, and quantitative analysis. • Design low-dose/fast-acquisition imaging algorithms using deep learning. • Develop physics-informed correction models for PET, CT, and MRI modalities. • Implement quantitative image analysis and multimodal registration pipelines. *Methodology:* Deep neural network training guided by physics-based modeling and iterative optimization using diverse clinical datasets. *Milestones:* • M1.1 (Month 12): First-generation algorithm prototypes completed. • M1.2 (Month 24): Algorithms meeting predefined quantitative benchmarks (e.g., >20% SNR gain, <5% quantitative bias). WP2. System Implementation and Optimization (Months 13–30) *Objective:* Translate algorithms into high-performance, clinically compatible software systems.

- Implement GPU-accelerated real-time processing and workflow integration.
- Optimize compatibility across major scanner vendors and imaging protocols.
- Conduct large-scale stress testing to ensure robustness and scalability.
  Methodology: Modular software architecture design, iterative performance tuning, and V&V testing under real-world data loads.
  Milestones:
- M2.1 (Month 18): Integrated prototype ready for internal testing.
- M2.2 (Month 30): System meets target processing speed (<1 min per volume) and operational stability (>99% uptime).

WP3. Clinical Validation and Benchmarking (Months 25–36)

Objective: Validate clinical reliability, generalizability, and regulatory readiness.

- Conduct large-scale retrospective validation using MGB's multi-ethnic datasets.
- Perform reader studies and cross-scanner performance benchmarking.
- Demonstrate clinical non-inferiority or superiority versus current practices.
  Methodology: Statistical evaluation of diagnostic accuracy (AUC, Dice, PPV/NPV), inter-reader agreement analysis, and longitudinal reproducibility studies.
  Milestones:
- M3.1 (Month 33): Clinical non-inferiority/superiority demonstrated.
- M3.2 (Month 36): Validation completed with FDA- and KFDA-grade documentation.

3. Timeline Overview (36 Months)

Period	Main Activities	Key Milestones
0–6 months	Planning, governance, pilot study	M0.1, M0.2
7–12 months	Algorithm development (Gen-1)	M1.1
13–24 months	Algorithm refinement & system integration	M1.2, M2.1
25–30 months	System optimization & large-scale testing	M2.2
31–36 months	Clinical validation & benchmarking	M3.1, M3.2

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Mass General Brigham (MGB), through the Center for Advanced Medical Computing and Analysis, will serve as the clinical and scientific lead in this international joint R&D project. MGB will contribute world-class expertise in medical imaging science, clinical workflows, and large-scale, data-driven research to ensure that all developed technologies are clinically relevant, scientifically rigorous, and globally competitive. 1. Technical and Scientific Contributions • Lead the definition of clinical requirements and target performance specifications for PET, CT, and MRI image generation, correction, and quantitative analysis. • Provide core expertise in advanced medical computing, AI-based image analysis, and physics-informed modeling. • Jointly develop and optimize novel imaging algorithms. • Participate in technical design reviews, algorithm evaluation, and performance benchmarking. 2. Research Scope and Leadership • Guide the overall research direction to align with real-world clinical needs and international standards. • Lead the pilot and feasibility study phase, including baseline benchmarking and proof-of-concept evaluation. • Define clinically meaningful validation endpoints and statistical methodologies for evaluation. • Contribute to the clinical interpretation and translational relevance of research outcomes. 3. Clinical Data and Research Resources • Provide access to large-scale, real-world PET/CT/MRI imaging datasets, within institutional and regulatory frameworks. • Support data analysis through secure, on-site and remote compute environments at MGB. • Supply expert-curated datasets, annotations, and reference standards where appropriate. • Facilitate access to heterogeneous imaging data across multiple scanners, acquisition protocols, and disease domains. 4. Clinical Validation and Evaluation • Lead or co-lead retrospective clinical validation studies using MGB datasets. • Conduct reader studies, quantitative benchmarking, and reproducibility assessments. • Evaluate diagnostic reliability, clinical usability, and quantitative accuracy of developed technologies. • Provide validation reports and supporting documentation suitable for regulatory submission and commercialization planning. 5. Infrastructure and Research Environment • Provide access to high-performance computing infrastructure and secure data environments aligned with clinical data governance standards. • Support collaborative R&D through regular technical workshops and virtual meetings. • Designate principal investigators, research fellows, and technical staff to participate in joint research activities. 6. Collaboration, Knowledge Transfer, and Dissemination • Actively participate in joint technical committees and milestone review meetings. • Contribute to joint intellectual property generation, subject to agreed governance and IP

	frameworks. • Co-author scientific publications and conference presentations in leading international venues. • Facilitate knowledge transfer and technical training to Korean SMEs to enable system implementation and commercialization. 7. Scope Limitations and Coordination • MGB's responsibilities will focus on research, algorithm development, and clinical validation, while product development, system engineering, and commercialization will be led by Korean SMEs. • All collaborative activities will be conducted in compliance with institutional policies, ethical guidelines, and applicable regulatory requirements.
Expected R&R of Korean SME Partner	**1. Technical Contributions and Core Development** • Lead the implementation and engineering of imaging algorithms jointly developed with Mass General Brigham (MGB) • Translate research-grade models into robust, scalable, and efficient software modules suitable for clinical environments • Optimize algorithms for real-time or near–real-time performance using GPU acceleration and advanced computing techniques • Design modular and extensible system architectures enabling integration with existing imaging devices and clinical software **2. System Development and Integration** • Develop prototype systems integrating: image reconstruction, correction, quantitative analysis modules • Ensure interoperability with multiple scanner vendors, imaging protocols, and hospital IT systems • Integrate developed technologies into imaging workstations, reconstruction engines, or AI-assisted clinical software platforms • Implement user interfaces and workflow-compatible tools in collaboration with clinical partners **3. Testing, Verification, and Validation Activities** • Conduct functional testing and performance verification using internal datasets and simulated imaging data • Collaborate with MGB on clinical validation by preparing test pipelines and supporting large-scale retrospective and prospective evaluations ○ Perform robustness testing across imaging modalities, scanner types, and acquisition protocols • Document test results to support regulatory and commercialization requirements **4. Resource Commitments** • Commit dedicated engineering and research personnel, including software engineers, imaging scientists, and AI specialists • Provide development infrastructure such as high-performance computing resources and software development and testing environments • Allocate internal R&D funding and in-kind contributions to support system development and testing **5. Commercialization and Regulatory Preparation** • Lead product planning, roadmap definition, and market analysis • Prepare documentation aligned with medical device software regulations (e.g., MFDS, FDA) • Define post-project commercialization strategies • **6. Collaboration and Project Management** • Actively participate in joint R&D governance structures, including technical committees and milestone reviews • Coordinate closely with MGB on algorithm refinement, validation strategy, and interpretation of clinical results

	• Facilitate knowledge transfer from research outputs to product development • Contribute to joint IP creation, documentation, and dissemination activities

5. Expected Outcomes/Impact

Expected Outcomes and Impact	This international joint R&D project is expected to deliver high-impact technological, commercial, and industrial outcomes by combining the clinical and scientific leadership of Mass General Brigham (MGB) with the engineering and commercialization capabilities of Korean small and medium-sized enterprises (SMEs).
	
	1. Technological Outcomes
	• Development of clinically validated, AI-enabled medical imaging technologies for PET, CT, and MRI, including:
	  ○ Low-dose and fast-acquisition image generation
	  ○ Physics-informed image correction and quantitative analysis
	  ○ Multi-modality data integration and clinical decision support
	• Demonstrated quantitative improvements in image quality, accuracy, and reproducibility, validated using large-scale, real-world clinical data from MGB
	• Establishment of a standardized clinical validation framework applicable across scanners, institutions, and imaging protocols
	• Advancement of technology readiness levels (TRLs) from concept and prototype stages to clinically deployable systems
	
	2. Commercialization and Market Impact
	• Creation of commercialization-ready prototype systems suitable for integration into:
	  ○ Medical imaging scanners
	  ○ Reconstruction workstations
	  ○ AI-assisted clinical software platforms
	• Strengthened global market competitiveness of Korean SMEs through:
	  ○ Validation in a world-leading clinical environment
	  ○ International co-development with recognized clinical experts
	• Accelerated pathways toward regulatory approval (MFDS, FDA) and global market entry
	• Generation of high-value intellectual property and joint publications supporting technology differentiation and branding
	
	3. Industrial and Economic Impact
	• Enhancement of the global competitiveness of the Korean medical imaging industry, particularly in AI-based and quantitative imaging technologies
	• Creation of new business opportunities and export potential for Korean SMEs in global healthcare markets
	• Contribution to high-value job creation in medical imaging, AI, and digital healthcare sectors
	• Strengthening of international R&D collaboration models linking global clinical leaders with Korean industry
	
	4. Healthcare and Societal Impact
	• Improved clinical outcomes through:
	  ○ Reduced radiation dose and scan time
	  ○ More accurate and reliable diagnostic and quantitative imaging
	• Support for precision medicine and theranostics, enabling patient-specific treatment planning and monitoring
	• Potential reduction in healthcare costs through more efficient imaging workflows and improved diagnostic accuracy
	
	5. Long-Term Strategic Impact
	• Establishment of a sustainable international joint R&D platform for future collaborative projects
	• Positioning Korea as a key contributor to next-generation medical imaging innovation
	• Expansion of long-term partnerships between Korean SMEs and global medical institutions